

Theory Assignment 3- 2025-2026
Introduction to Databases (CSE 3151)
Programme: B.Tech. (CSE & CSIT) Semester: 6th
Submission Date: 16-05-2025

Course Learning Outcome	*Taxonomy Level
Interpret the functional issues related to transaction and database recovery along with the concept of storage and database system architecture.	L4

This assignment is designed to give you practice with the concepts of

- Transaction management
- Database recovery

1. Check whether the given schedules are conflict serializable or not-

a. S1: r1(x)r1(y)r2(x)r2(y)w2(y)w1(x)

b. S2: r1(x)r2(x)r2(y)w2(y)r1(y)w1(x)

Let $r_i(z)$ and $w_i(z)$ denote read and write operations respectively on a data item z by a transaction T_i . Consider the following two schedules.

2. Check whether the given schedule S is conflict serializable and recoverable or not.

T1	T2	T3	T4
	READ(X)		
		WRITE(X) COMMIT	
WRITE(X) COMMIT			
	WRITE(Y) READ(Z) COMMIT		
			READ(X) READ(Y) COMMIT

Schedule S

3. Consider a schedule of transaction T1 and T2:

T1	RA			RC		WD		WB	Commit	
T2		RB	WB		RD		WC			Commit

Here, RX stands for “Read(X)” and WX stands for “Write(X)”. Which one of the following schedules is conflict equivalent to the above schedule?

(A)

T1	RA	RC	WD				WB		Commit	
T2				RB	WB	RD		WC		Commit

(B)

T1				RA	RC	WD	WB		Commit	
T2	RB	WB	RD					WC		Commit

4. Consider the following two transactions:

T1: **read(A);**
 read(B);
 if A = 0 then B := B + 1;
 write(B).
T2: **read(B);**
 read(A);
 if B = 0 then A := A + 1;
 write(A).

Add lock and unlock instructions to transactions T1 and T2, so that they observe the two-phase locking protocol. Can the execution of these transactions result in a deadlock?

5. Check whether the given schedule below can complete all its operations in the given order using timestamp based protocol or not.

T1	T2
READ(A)	
	READ(A)
WRITE (A)	
	WRITE(A)

6. Consider the following log sequence:

<T1 start> <T1, A, 100, 150> <T2 start> <T2, B, 200, 250> <T1 commit> <crash>

- a) Identify which transactions need UNDO and REDO.
- b) Show the final values of A and B after recovery.

*****End*****